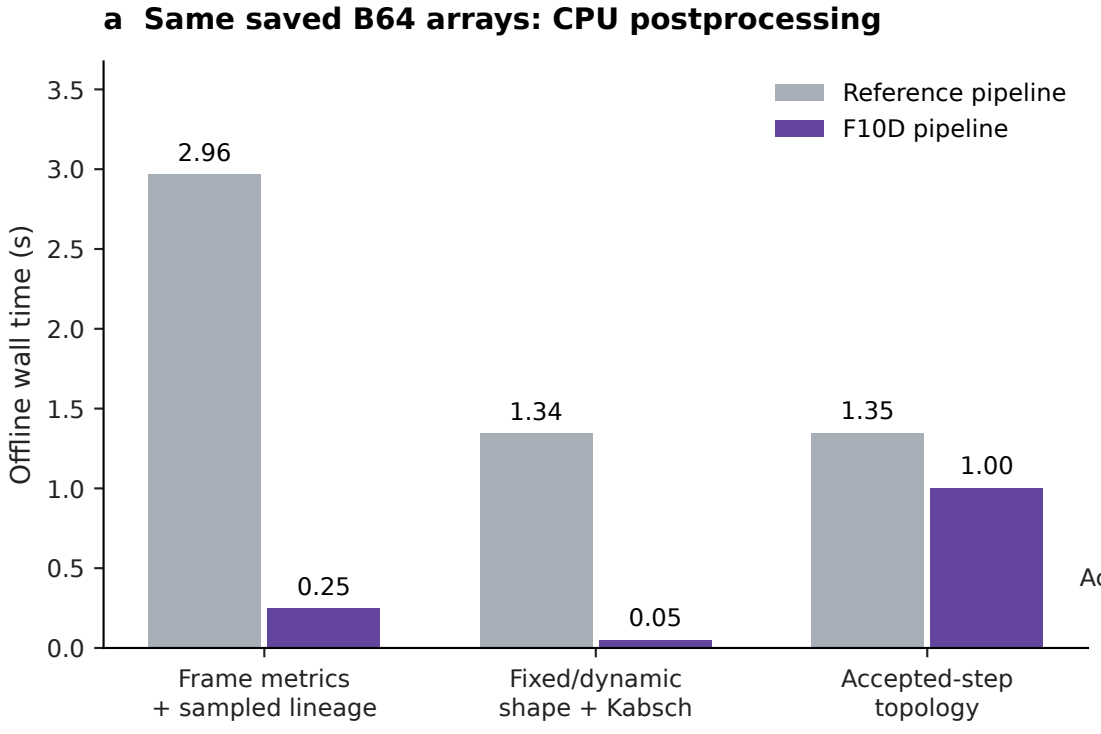

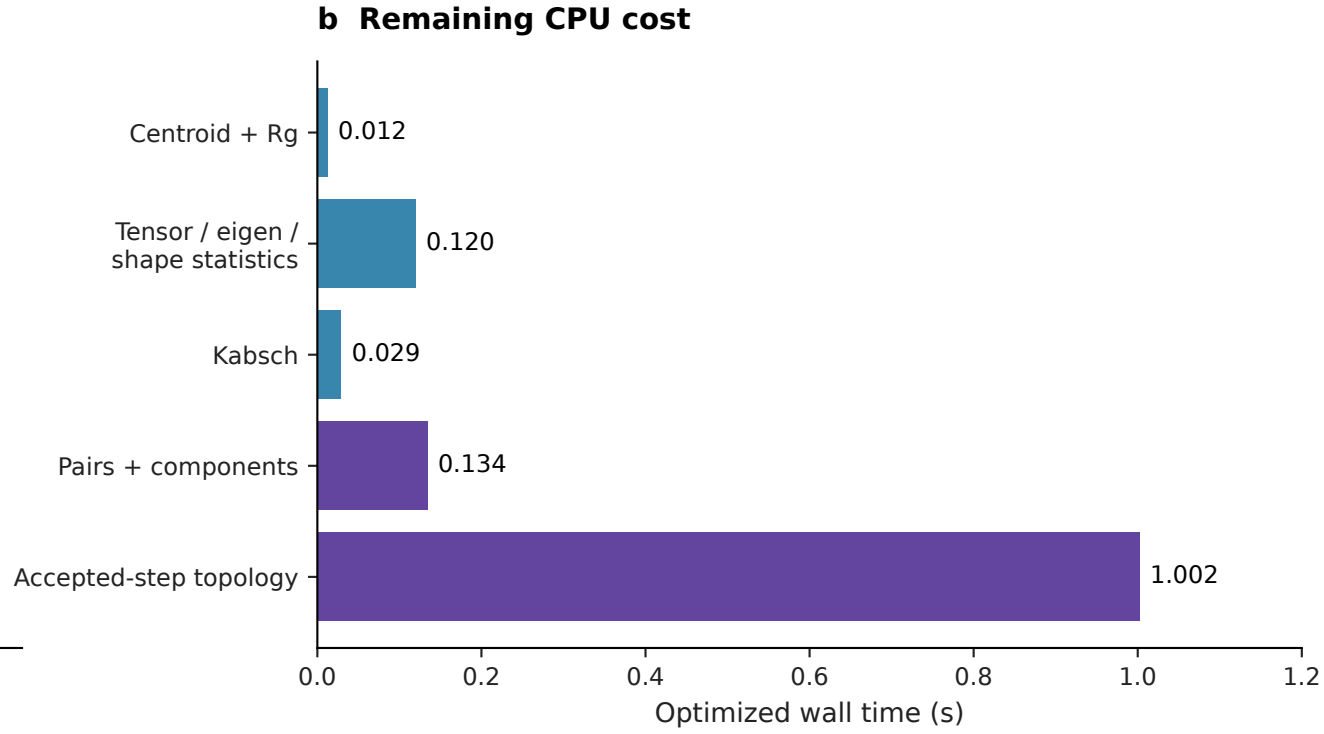

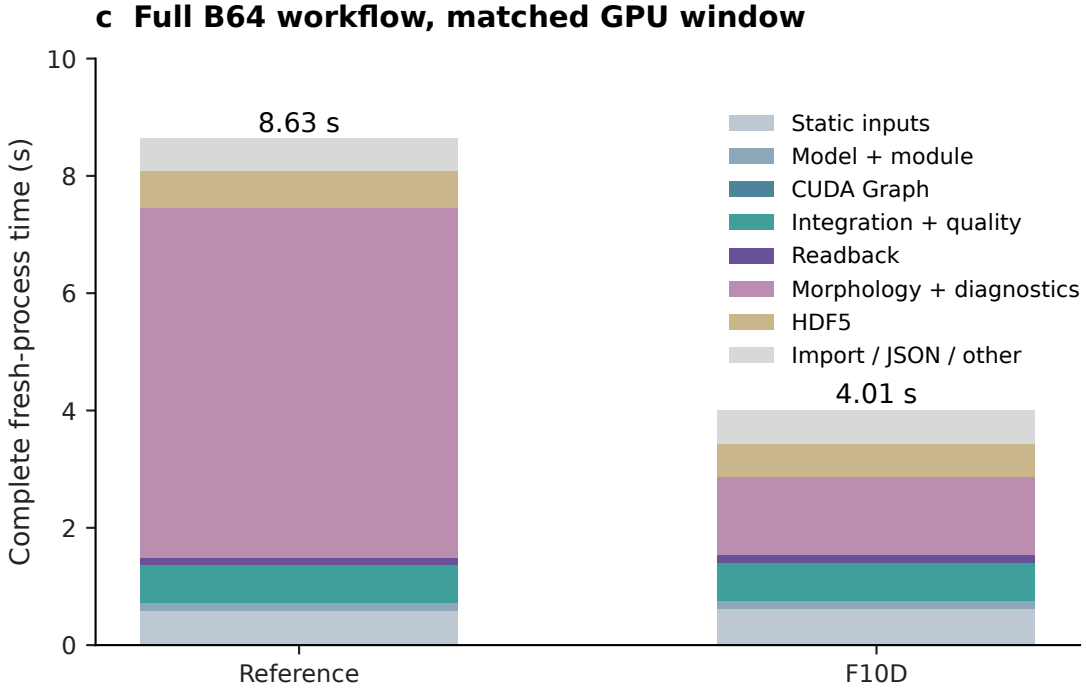

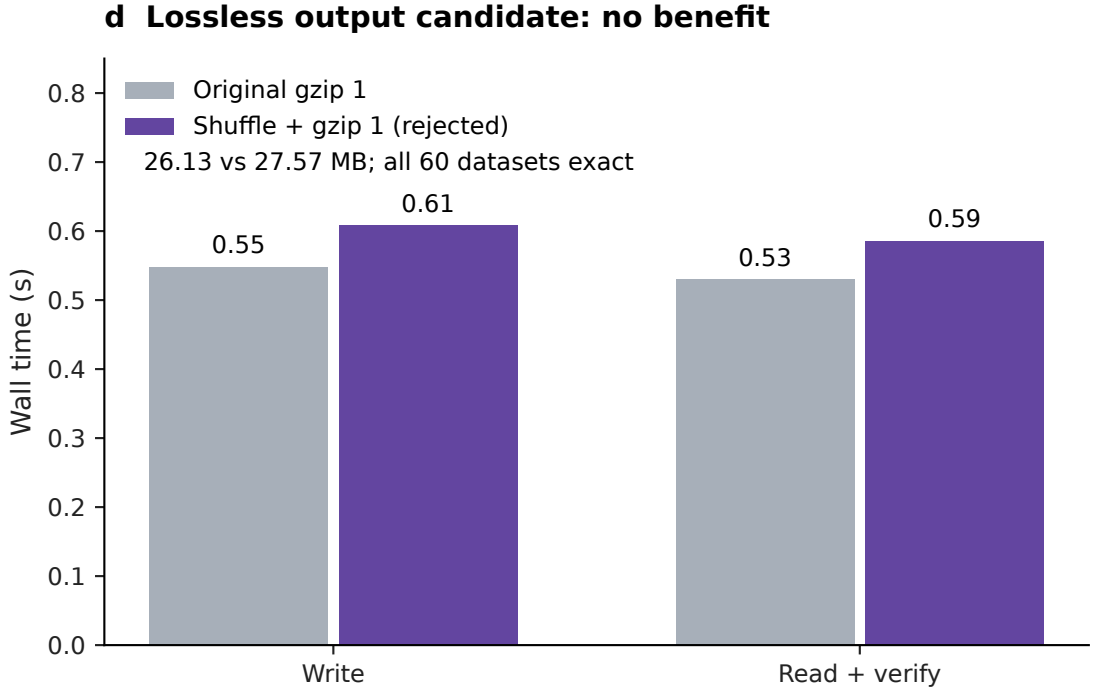

B64 = 64 x 32 bubbles; 10 ms, 201 frames. Reference centroid/Rg subcosts were not isolated.
CPU timings exclude imports; cold total includes them. Final pipeline retains original gzip 1. No quality checks removed.